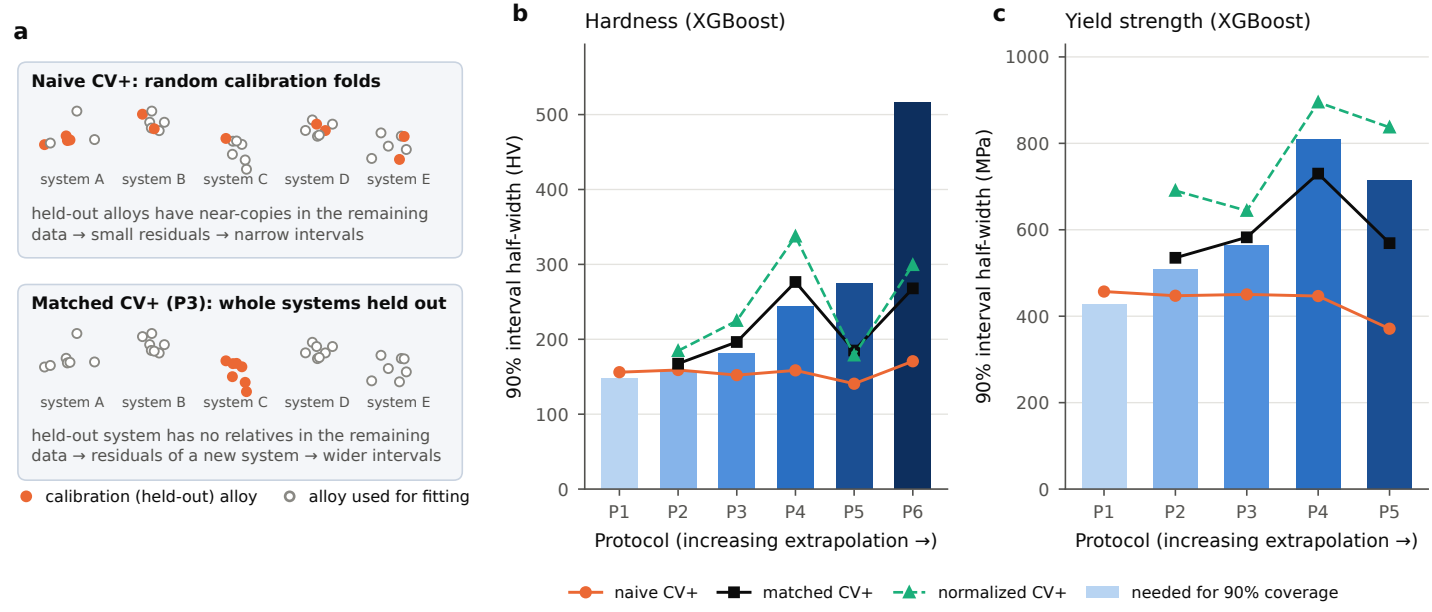
a
Naive CV+: random calibration folds
system A
system B
system C
system D
system E
held-out alloys have near-copies in the remaining data → small residuals → narrow intervals
Matched CV+ (P3): whole systems held out
system A
system B
system C
system D
system E
held-out system has no relatives in the remaining data → residuals of a new system → wider intervals
calibration (held-out) alloy
alloy used for fitting
b
Hardness (XGBoost)
90% interval half-width (HV)
0
100
200
300
400
500
P1
P2
P3
P4
P5
P6
Protocol (increasing extrapolation →)
c
Yield strength (XGBoost)
90% interval half-width (MPa)
0
200
400
600
800
1000
P1
P2
P3
P4
P5
Protocol (increasing extrapolation →)
naive CV+
matched CV+
normalized CV+
needed for 90% coverage